

Supplemental Material: TBMaLT, a flexible toolkit for combining tight-binding and machine learning

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S 1. COMPARISON OF VARIOUS STRATEGIES TO TRACK THE GRADIENTS DURING SELF-CONSISTENCY CYCLES

We have investigated, whether the choice of the gradient tracking process makes a significant impact on the prediction results. Two different gradient tracking strategies were used in this study. In the first one, the gradient was tracked through all cycles while the self-consistent charges (SCC) were obtained. For the second strategy, the SCC cycles were performed outside of the gradient tracking process at first. Once self-consistency was achieved, the converged charges were used as initial input for a single diagonalization, which was carried out with gradient tracking.

The same rattled diamond type silicon systems as shown in the main text (Section IV.A.2) and the same training method using optimized splines for the diatomic integrals was used in this case. To simplify the training, we have used the same training set for both strategies, and it consisted of a single system only. The performances of the models using two different gradient tracking strategies are summarized in Figure S1. As shown in Figure S1 (a), the decays of the loss functions are nearly identical for both strategies. By testing on 20 unseen systems, the prediction results of the two models show little differences only (Figure S1 (c) and (d)), and both are considerably better as the DFTB calculations using the `siband-1-1` parametrization set (Figure S1 (b)), which was used as the starting point of the optimization.

It should be noted, that the system investigated here is mostly homogeneous, where the charge transfer between the atoms is rather small. Consequently, the values of the converged charges at the end of the SCC-procedure are expected to depend only weakly on the values of the input feature vector. In such cases, the two different strategies deliver very similar results, as demonstrated above. For more complex systems with significant charge transfer, however, the impact of the gradient tracking is expected to be much more pronounced, and the strategy with the single diagonalization, which yields approximative gradients only, might even lead to wrong results. We plan to investigate this question more systematically in the future.

S 2. MACHINE LEARNING SETTINGS

The atom-centered symmetry functions (ACSFs)¹ have been implemented in TBMaLT. In Section IV.A.1 of the main text, G_1 , G_2 and G_4 have been used and the parameters of G_1 , G_2 and G_4 were [6.0], [0.5, 1.0] and [0.02, 1.0, -1.0], respectively. All symmetry functions are atom-wise

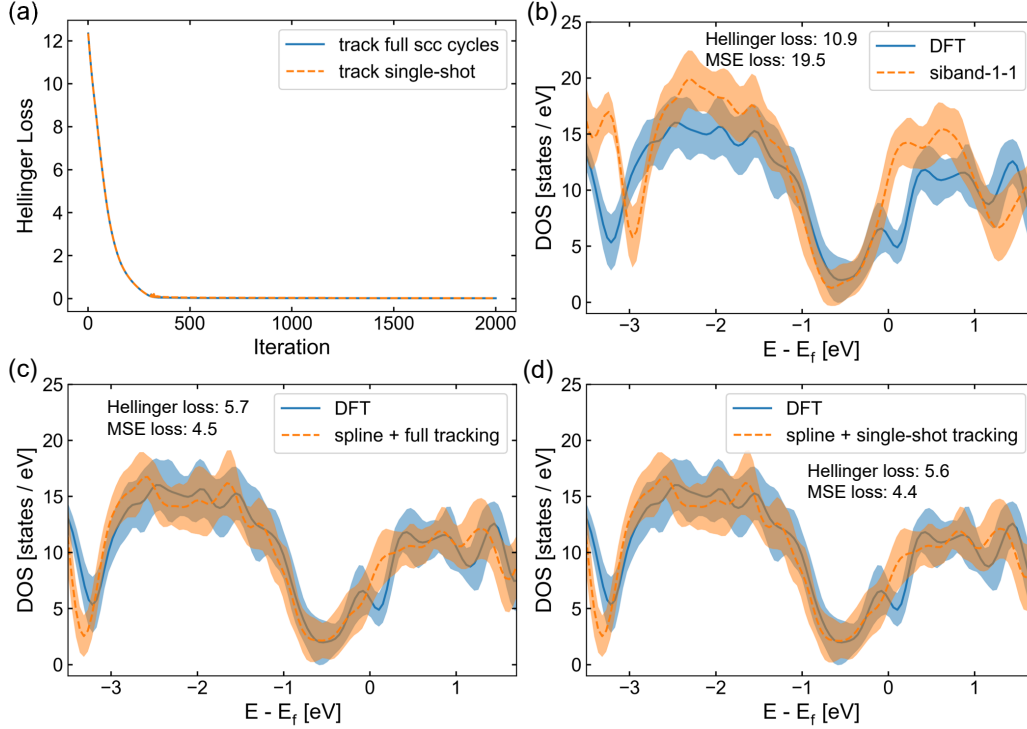


FIG. S 1. Comparison between two gradient tracking strategies. (a) Hellinger loss vs. training iteration when using different gradient tracking strategies, (b) the DOS curves of the test systems calculated by DFT and DFTB using the `siband-1-1` parameter set, (c) comparison of the DOS curves for the test systems with DFT references when using full-SCC gradient tracking to optimize splines of the diatomic integrals and (d) comparison of the DOS curves for the test systems with DFT references when using external SCC cycles and a single final diagonalization with gradient tracking to optimize splines of the diatomic integrals.

representations.

The random forest predictions have been performed by using Scikit-learn². The number of estimators in random forest regression was 100, and for all other parameters used for the predictions the default settings had been used.

REFERENCES

- ¹J. Behler, “Atom-centered symmetry functions for constructing high-dimensional neural network potentials,” *J. Chem. Phys.* **134**, 074106 (2011).
- ²F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, *et al.*, “Scikit-learn: Machine learning in python,” *J. Mach.*

Learn. Res. **12**, 2825–2830 (2011).